

Clauses

Clause order: SELECT → FROM → WHERE → GROUP BY → HAVING → ORDER BY → LIMIT

SELECT x	Selects the column x from the table.
SELECT *	Selects all columns from the table.
SELECT DISTINCT x	Selects unique values from the column x.
FROM table	Specifies the table to select from.
WHERE x	Filters the results based on the condition x.
ORDER BY x	Sorts the results based on the column x.
GROUP BY x	Groups the results based on the column x.
AS x	Renames a column or table to x.
HAVING x	Filters the results of a GROUP BY clause based on the condition x.
LIKE x	Used in WHERE clauses to match a pattern. "% " many characters, "_" single character
LIMIT x	Limits the number of rows returned to x.

Table Modifiers

It is a good practice to Select before modifying a table.

INSERT INTO x(cols) VALUES (1st)	Make a row-wise insertion of the values in 1st into the table x.
CREATE TABLE x (cols)	Creates a new table named x with the
DELETE FROM x WHERE c	Deletes rows from the table x where the c is met.
DROP TABLE x	Deletes the table x.
UPDATE x SET c=v WHERE c	Updates the column c in table x to the value v if the c is met.

In create queries we can specify the data type and constraints of each column. The most common data types are:

INTEGER	A signed integer.
REAL	A floating point value.
TEXT	A text string.
BLOB	A binary large object.

And the most common constraints are:

PRIMARY KEY	Uniquely identifies each row in the table.
FOREIGN KEY	Establishes a relationship between two tables.
NOT NULL	Ensures that a column cannot have a NULL value.
UNIQUE	Ensures that all values in a column are unique.
CHECK	Ensures that all values in a column satisfy a specific condition.
DEFAULT	Sets a default value for a column when no value is specified.

Operator

LIKE x	Used in WHERE clauses to match a pattern. % represents zero, one, or multiple characters. _ represents a single character.
AND x	Combines multiple conditions in a WHERE clause.
OR x	Combines multiple conditions in a WHERE clause.
NOT x	Negates a condition in a WHERE clause.
IN (x, y)	Checks if a value is in a list of values.
BETWEEN x AND y	Checks if a value is within a range of values.
IS NULL	Checks if a value is NULL.
IS NOT NULL	Checks if a value is not NULL.

Functions

TOP(n)	Limits the number of rows returned to n.
COUNT(x)	Counts the number of non-null values in column x.
SUM(x)	Calculates the sum of values in column x.
AVG(x)	Calculates the average of values in column x.
MIN(x)	Finds the minimum value in column x.
MAX(x)	Finds the maximum value in column x.

Examples

```
SELECT Orders.OrderID, Customers. FirstName,
Customers.LastName, Customers.Email, Orders.OrderDate
FROM Orders
JOIN Customers ON Orders.CustomerID=Customers.CustomerID
ORDER BY OrderID
```

Nested Queries

```
SELECT * FROM Orders
WHERE CustomerID=(
    SELECT CustomerID FROM Customers
    WHERE Email="ex@example.com")
```

Insert Queries

```
INSERT INTO EMPLOYEES(NAME, FIRSTNAME, DEPT,DEPTFLOOR)
VALUES ('Brown', 'Tom', 'Management', 1)
```

Table Creation

```
CREATE TABLE EMPLOYEES(
    "id" INTEGER NOT NULL,
    "name" TEXT NOT NULL,
    "first_name" TEXT NOT NULL,
    "dept_id" TEXT NOT NULL,
    "dept_floor" INTEGER NOT NULL,
    PRIMARY KEY("id"),
    FOREIGN KEY("dept")
        REFERENCES "Departments"("dept_id")
)
```